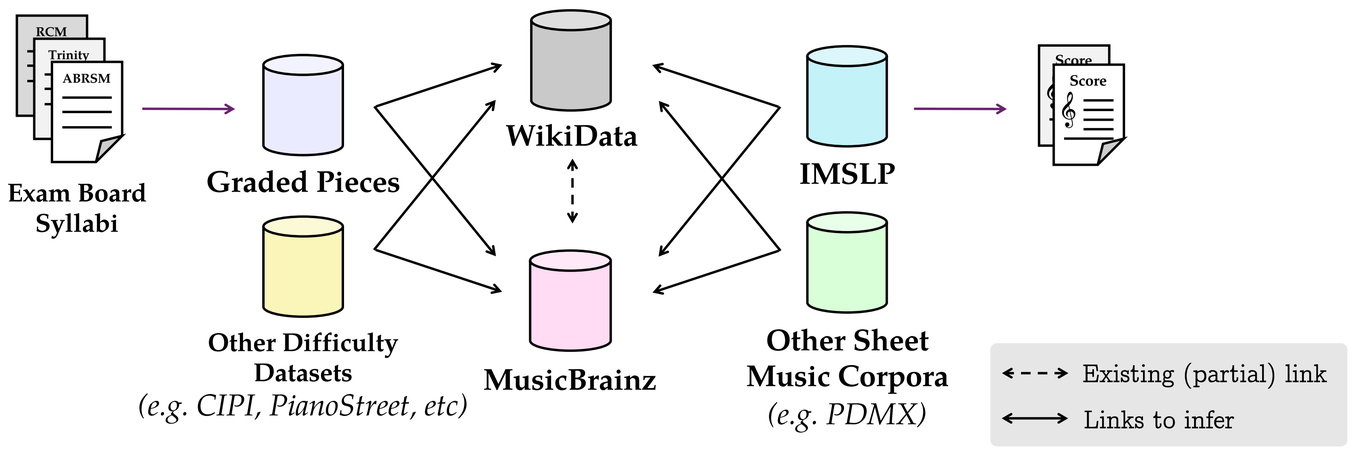
MetaMatch: Reconciling Metadata for Aggregating Music Difficulty Annotations

Megan E. Finch[1], Chris G. Willcocks[1], Mark R. H. Gotham[2]
[1] Durham University, UK [2] King's College London, UK

Motivation

- Recent work in automated performance difficulty recognition focuses predominantly on piano music, due to data availability.
- Music exam board repertoire lists provide a rich source of difficulty labels for dozens of instruments; and the sheet music for many of the listed works is in the public domain.
- To make use of this data, we must first link musical works across multiple corpora with inconsistent metadata.

Application: Aggregating Music Difficulty Annotations

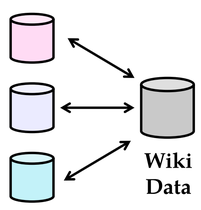


Dataset

- **58,104 repertoire list entries** collated from exam boards, covering **20+ instruments** and **13k+ composers**.
- Includes **7,737 entries** for piano.

Previous work

- The **LinkedMusic** project [1] uses OpenRefine's reconciliation API to semi-automatically match as many values to WikiData entities as possible.



- Peeters & Bizer [2] showed that LLMs with little or no fine-tuning perform competitively on entity matching tasks.

(Some) Challenges

Ambiguous composers

Query: "Jean-Baptiste Loeillet" or "Loeillet"

WikiData	
ID	**Label**
Q1319135	Jean-Baptiste Loeillet of London
Q1347001	Jean Baptiste Loeillet of Ghent

Composer names can be under-specified or ambiguous.

Alternative titles

Graded Pieces	
Title	Chaconne
Composer	Handel
Book	*Classics to Moderns*, Book 4
Difficulty	MTB, Grade 4, Piano

IMSLP	*Match?*
Suite in G minor, HWV 432	✓
Chaconne in F major, HWV 485	✗
Chaconne in G major, HWV 435	✗
Chaconne in G major, HWV 442	✗

Pieces can have different titles, including **translations and levels of detail**. For example, *Chaconne* here refers to movement *VI. Passacaglia* of HWV 432.

Different granularities

Graded Pieces	
Title	In Yugoslav Mode
Composer	Béla Bartók
Book	*MTB Piano Grade 1*
Difficulty	Grade 1

is part of → **IMSLP**: Mikrokosmos, Sz. 107

is the same piece as → **MusicBrainz**: Mikrokosmos, Sz. 107, BB 105: Volume II, No. 40: In Yugoslavian Mode (Allegretto)

No consistent meaning of 'musical work' across database schemas.

Missing entities

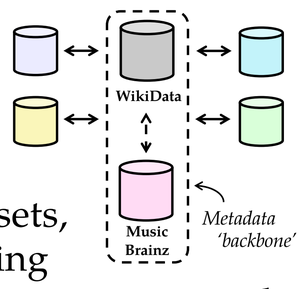
Query: "Tom Davoren"

WikiData: No entries

MusicBrainz.Artists	
ID	**Name**
d8acb7b0…	Tom Davoren
527f3f97-7…	Tony Davoren
57c6f649…	Tom Jones

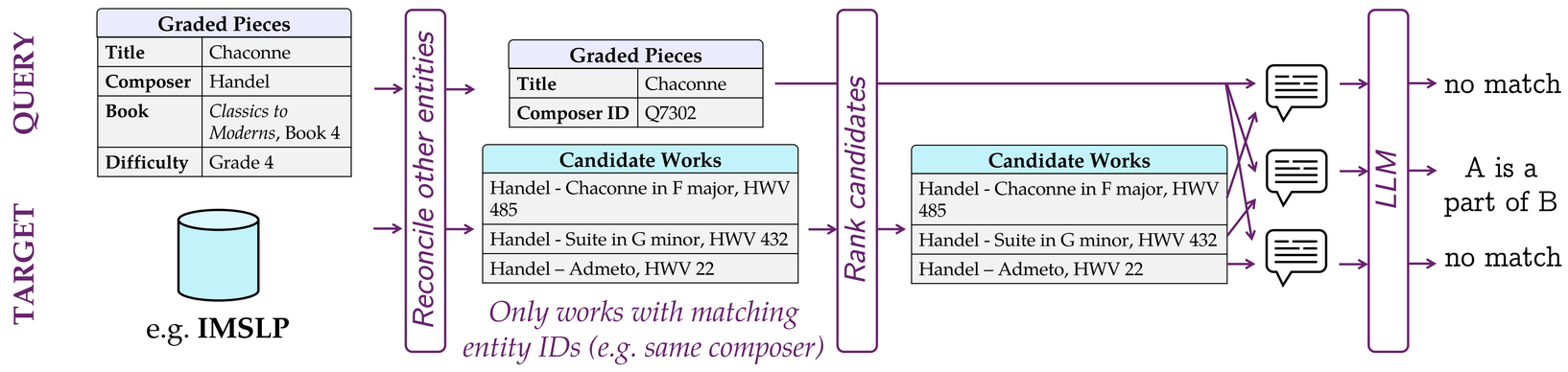
Many musical works, and even artists, have no WikiData entry; **no single 'backbone' dataset covers all entities** one might wish to link.

Method



- **Metadata backbone:** attempt to find a more complete matching by routing through *any* of several 'backbone' datasets, and by leveraging existing links between them as a weak supervision signal. Matching against two target datasets which have such a link (e.g. *some* MusicBrainz works link to an IMSLP URL) allows us to assess matching accuracy/consistency.
- **Match composers:** for tractability, we first filter candidate works for each graded piece entry using *other* entities which are easier to reconcile (e.g. composers).
- **LLM-based entity recognition:** ask the LLM to classify (query, candidate) pairs as 'match', 'no match', or 'piece A is a part of piece B' (or vice versa).
- **Evaluation dataset:** to assess LLM accuracy, we manually label 447 graded pieces with their closest match in MusicBrainz, where available.

Matching Pipeline



Results

How many pieces can be matched?

Of 5,057 piano pieces with known composers:.

	MB.Works		IMSLP.Works	
No match	1,753	(34.7%)	2,668	(52.8%)
Any match	3,304	(65.3%)	2,389	(47.2%)
Exact	2,331	(46.1%)	954	(18.9%)
Part of	782	(15.5%)	1,427	(28.2%)
Contains	191	(3.8%)	8	(0.2%)

47% have a matching IMSLP work.

How many composers can be matched?

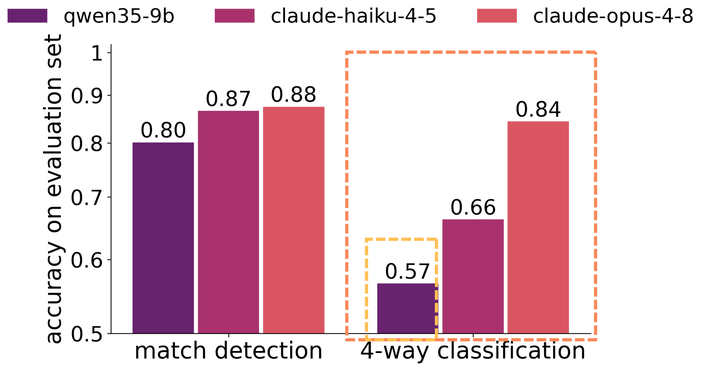
	GP.Composers	
Size	13,583	
Matched to WD	6,624	(48.8%)
Matched to MB	11,169	(82.2%)
Matched to both	6,116	
No MB→WD link	1,937	
Link agrees	3,615	(86.5%)
Link disagrees	564	(13.5%)

11k composers matched to a MusicBrainz artist.

Matching to MusicBrainz vs WikiData is **largely consistent.**

How accurate is the LLM matching?

LLMs can accurately detect when two pieces match (i.e. 'match' vs 'no match'), but **not the type of match** (i.e. 'match' vs 'A is part of B').



Commercial models substantially **outperform SOTA open weight models.**

Conclusion

- Present method for reconciling entities across heterogenous music datasets, combining existing reconciliation services with LLM-based entity matching.
- Demonstrate that this method can be used to link entries from **music exam board repertoire lists to IMSLP works**, and hence **produce new difficulty-labelled datasets for multiple instruments.**

References

1. I. Fujinaga, "Sustainable archiving of music databases through RDF and NLQ2SPARQL," in *Proceedings of the 12th International Conference on Digital Libraries for Musicology*, 2025, pp. 100–104.
2. R. Peeters and C. Bizer, "Using ChatGPT for entity matching," in *European Conference on Advances in Databases and Information Systems*. Springer, 2023, pp. 221–230.